human

Which buildings in Exeter are within 250m of an electricity pylon?

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to find which buildings in Exeter are within 250m of an electricity pylon. Provide the general steps to solve this geospatial query.'}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings in Exeter, and we need to find electricity pylons in Exeter as well. The condition is to identify buildings that are within 250m of an electricity pylon.

output steps: ["Find Exeter 1 area", "Find buildings in Exeter as many search results", "Find electricity pylons in Exeter as many search results", "Apply condition buildings within 250m of electricity pylons"]

host_agent

Tool send_message returned result : The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings in Exeter, and we need to find electricity pylons in Exeter as well. The condition is to identify buildings that are within 250m of an electricity pylon.

output steps: ["Find Exeter 1 area", "Find buildings in Exeter as many search results", "Find electricity pylons in Exeter as many search results", "Apply condition buildings within 250m of electricity pylons"]

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..,\n count of records fetched : 98.\n ',] Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    # Check for 'Exeter' in name1_text, and also ensure it's a city or settlement
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter'))) &
                   (gdf['description'].str.lower().isin(['city', 'town', 'suburban area', 'part of settlement', 'village', 'settlement']))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = "GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement-related descriptions."
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered to 1 polygon(s) for Exeter using name and description columns.',]

Filtered to 1 polygon for Exeter using the name and description columns. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) representing Exeter, filtered by relevant settlement-related descriptions. If you need further details or analysis on this filtered polygon, please let me know!

named_area_agent

The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) representing Exeter, filtered by name and settlement-related descriptions.

host_agent

Calling tool send_message with args : {'target': 'buildings', 'task_description': "Find all buildings within the area defined by 'filtered_exeter_area_polygon'."}

buildings_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'all_buildings_in_filtered_exeter_area'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : [\'buildings_all_buildings_in_filtered_exeter_area\', \'buildingline_all_buildings_in_filtered_exeter_area\', \'buildingpart_all_buildings_in_filtered_exeter_area\'],\n Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find all buildings within the area defined by 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Find all buildings within the area defined by 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Find all buildings within the area defined by 'filtered_exeter_area_polygon'. using filters None."],\n counts of records fetched are : [69169, 69342, 85480].\n ', [, ,]]

The search for all buildings within the area defined by 'filtered_exeter_area_polygon' has returned the following data artifacts:

1. buildings_all_buildings_in_filtered_exeter_area: Contains 69,169 records of building data within the specified area.
2. buildingline_all_buildings_in_filtered_exeter_area: Contains 69,342 records of building line data.
3. buildingpart_all_buildings_in_filtered_exeter_area: Contains 85,480 records of building part data.

If you need a specific type of building data (e.g., whole buildings, building lines, or building parts), please specify which artifact you would like to use for further analysis or download.

Summary of results: - Artifact names: buildings_all_buildings_in_filtered_exeter_area, buildingline_all_buildings_in_filtered_exeter_area, buildingpart_all_buildings_in_filtered_exeter_area - Number of results: 69,169 (buildings), 69,342 (building lines), 85,480 (building parts)

Let me know if you need further filtering or analysis on any of these datasets!

host_agent

Calling tool send_message with args : {'target': 'structures_agent', 'task_description': "Find all electricity pylons within the area defined by 'filtered_exeter_area_polygon'."}

structures_agent

Calling tool call_os_ngd with args : {'filters': ['Electricity Pylon'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'electricity_pylons_exeter'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : [\'electricity_pylons_exeter\'],\n Descriptions are : ["A geopandas dataframe containing structure data with filters and bbox applied for the query Find all electricity pylons within the area defined by 'filtered_exeter_area_polygon'. using filters ['Electricity Pylon']. Further Name filtering is available for this"],\n counts of records fetched are : [19].\n ', []]

I found 19 electricity pylons within the area defined by 'filtered_exeter_area_polygon'. The data is saved in the artifact named 'electricity_pylons_exeter'.

If you need further analysis or filtering, please let me know!

host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot all buildings from 'buildings_all_buildings_in_filtered_exeter_area' that are within 250m of any electricity pylon from 'electricity_pylons_exeter'."}

plotting_agent

Tool Code Executor

```
def plot_buildings_within_250m_of_pylons(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import Polygon
    from shapely.ops import unary_union

    # Unpack data
    buildings = data[0]
    pylons = data[1]

    # Ensure both are in EPSG:4326 for folium
    buildings = buildings.to_crs(epsg=4326)
    pylons = pylons.to_crs(epsg=4326)

    # Create a 250m buffer around each pylon (convert to meters first, then back to 4326 for folium)
    pylons_m = pylons.to_crs(epsg=3857)
    pylons_m['buffer_250m'] = pylons_m.geometry.buffer(250)
    buffer_union = unary_union(pylons_m['buffer_250m'])
    buffer_gdf = gpd.GeoDataFrame(geometry=[buffer_union], crs=pylons_m.crs).to_crs(epsg=4326)

    # Find buildings that intersect the buffer
    buildings_in_buffer = buildings[buildings.intersects(buffer_gdf.iloc[0].geometry)]
```

```
    # Prepare folium map
    if not buildings_in_buffer.empty:
        centroid = buildings_in_buffer.geometry.iloc[0].centroid
        m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)
    else:
        centroid = pylons.geometry.iloc[0].centroid
        m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

    # Plot pylons
    folium.GeoJson(pylons[['geometry']], name='Electricity Pylons', style_function=lambda x: {'color': 'red', 'weight': 2}).add_to(m)
    # Plot buffer
    folium.GeoJson(buffer_gdf, name='250m Buffer', style_function=lambda x: {'color': 'blue', 'weight': 2, 'fillOpacity': 0.1}).add_to(m)
    # Plot buildings
    folium.GeoJson(buildings_in_buffer[['geometry']], name='Buildings within 250m', style_function=lambda x: {'color': 'green', 'weight': 1, 'fillOpacity': 0.3

    folium.LayerControl().add_to(m)
    map_filename = 'buildings_within_250m_of_pylons_map.html'
    m.save(map_filename)

    # Prepare summary
    summary_rows = buildings_in_buffer[['osid', 'description', 'buildinguse', 'geometry_area_m2']].head(5).to_dict('records')
    summary = f"Found {len(buildings_in_buffer)} buildings within 250m of any electricity pylon. First 5: {summary_rows}"
    artifact_name = 'buildings_within_250m_of_pylons_map'
    artifact_description = 'A folium map showing all buildings within 250m of any electricity pylon in Exeter, with pylons and buffer visualized.'
    return [summary, artifact_name, artifact_description, map_filename]
```

Tool code_executor returned result : ["Found 995 buildings within 250m of any electricity pylon. First 5: [{'osid': '32ff414d-d1c8-454b-9b67-ac472217e4ce', 'description': 'Domestic Outbuilding', 'buildinguse': 'Unknown', 'geometry_area_m2': 18.438}, {'osid': 'b9a49c80-fecf-4308-a98d-cf90ea4ea2e4', 'description': 'End-Of-Terrace House', 'buildinguse': 'Residential Accommodation', 'geometry_area_m2': 99.338}, {'osid': '9dfb4b41-f53d-44c0-8a7a-5f7884375677', 'description': 'Mid-Terrace House', 'buildinguse': 'Residential Accommodation', 'geometry_area_m2': 78.565}, {'osid': 'aea058f2-6a39-4829-ae6b-85c6863a250e', 'description': 'Mid-Terrace House', 'buildinguse': 'Residential Accommodation', 'geometry_area_m2': 66.91}, {'osid': 'ef3285ad-8a1d-4d4d-aa50-333888085e10', 'description': 'Unknown Building', 'buildinguse': 'Unknown', 'geometry_area_m2': 7.858}]",]

Found 995 buildings within 250m of any electricity pylon. First 5: [{'osid': '32ff414d-d1c8-454b-9b67-ac472217e4ce', 'description': 'Domestic Outbuilding', 'buildinguse': 'Unknown', 'geometry_area_m2': 18.438}, {'osid': 'b9a49c80-fecf-4308-a98d-cf90ea4ea2e4', 'description': 'End-Of-Terrace House', 'buildinguse': 'Residential Accommodation', 'geometry_area_m2': 99.338}, {'osid': '9dfb4b41-f53d-44c0-8a7a-5f7884375677', 'description': 'Mid-Terrace House', 'buildinguse': 'Residential Accommodation', 'geometry_area_m2': 78.565}, {'osid': 'aea058f2-6a39-4829-ae6b-85c6863a250e', 'description': 'Mid-Terrace House', 'buildinguse': 'Residential Accommodation', 'geometry_area_m2': 66.91}, {'osid': 'ef3285ad-8a1d-4d4d-aa50-333888085e10', 'description': 'Unknown Building', 'buildinguse': 'Unknown', 'geometry_area_m2': 7.858}] Artifact name: buildings_within_250m_of_pylons_map Artifact description: A folium map showing all buildings within 250m of any electricity pylon in Exeter, with pylons and buffer visualized. Artifact data:

buildings_within_250m_of_pylons_map.html

host_agent

There are 995 buildings in Exeter that are within 250 meters of an electricity pylon. These include various types such as domestic outbuildings and residential houses. The analysis was performed using the Exeter area boundary, all buildings within that area, and all identified electricity pylons.

A map artifact has been generated that visualizes: - All buildings within 250m of any electricity pylon in Exeter - The pylons and their 250m buffer zones

If you need the map or a list of these buildings, let me know!